

Chapter Six Correlation and Regression

Correlation is a statistical method used to determine whether a linear relationship between variables exists. **Regression** is a statistical method used to describe the nature of the relationship between variables, that is, positive or negative, linear or nonlinear. The purpose of this chapter is to answer these questions statistically:

1. Are two or more variables linearly related?
2. If so, what is the strength of the relationship?
3. What type of relationship exists?
4. What kind of predictions can be made from the relationship?

To answer the first two questions, statisticians use a numerical measure to determine whether two or more variables are linearly related and to determine the strength of the relationship between or among the variables. This measure is called a *correlation coefficient*.

To answer the third question, you must ascertain what type of relationship exists. There are two types of relationships: *simple* and *multiple*. In a **simple relationship**, there are two variables—an **independent variable**, also called an explanatory variable or a predictor variable, and a **dependent variable**, also called a response variable. A simple relationship analysis is called *simple regression*, and there is one independent variable that is used to predict the dependent variable. In a **multiple relationship**, called *multiple regressions*, two or more independent variables are used to predict one dependent variable. Simple relationships can also be positive or negative. A **positive relationship** exists when both variables increase and decrease at the same time. In a **negative relationship**, as one variable increases, the other variable decreases, and vice versa. This chapter is concerned with *simple regression*.

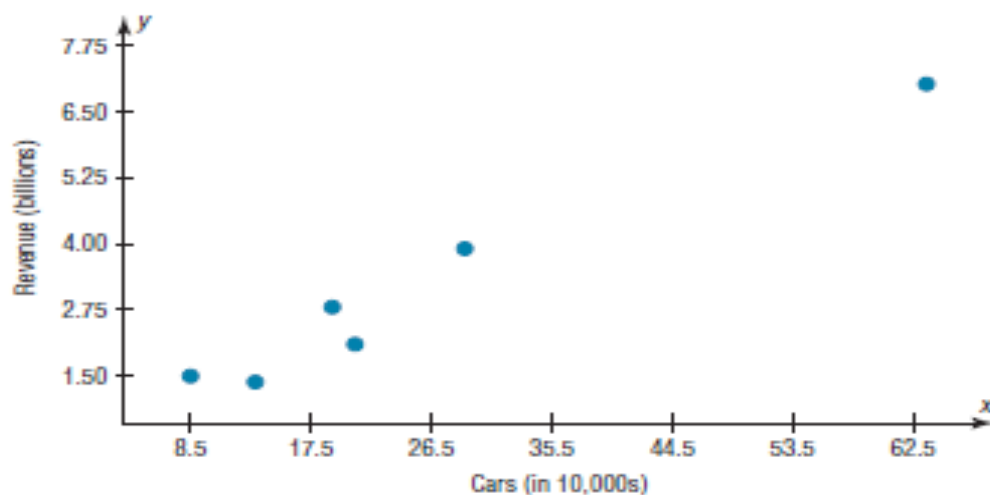
Finally, the fourth question asks what type of predictions can be made. Predictions are made in all areas and daily. That is, the stronger the relationship is between variables, the more accurate the prediction is.

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In simple correlation and regression studies, the researcher collects data on two numerical or quantitative variables to see whether a relationship exists between the variables. For example, if a researcher wishes to see whether there is a relationship between number of cars ready to be rented and its revenue, the researcher must select a random sample of companies, determine the number of cars in each company, and obtain their revenue. A table can be made for the data, as shown here.

Company	Cars (in ten thousands)	Revenue (in billions)
A	63.0	\$7.0
B	29.0	3.9
C	20.8	2.1
D	19.1	2.8
E	13.4	1.4
F	8.5	1.5

The independent and dependent variables can be plotted on a graph called a *scatter plot*. The independent variable x is plotted on the horizontal axis, and the dependent variable y is plotted on the vertical axis.



A **scatter plot** is a graph of the ordered pairs (x, y) of numbers consisting of the independent variable x and the dependent variable y . The scatter plot is a visual way to describe the nature of the relationship between the independent and dependent variables.

The **correlation coefficient** computed from the sample data measures the strength and direction of a linear relationship between two quantitative variables. The symbol for the

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sample correlation coefficient is r . The symbol for the population correlation coefficient is ρ (Greek letter rho).

The *range of the correlation coefficient* is from +1 to -1. If there is a *strong positive linear relationship* between the variables, the value of r will be close to +1. If there is a *strong negative linear relationship* between the variables, the value of r will be close to -1. When there is no linear relationship between the variables or only a weak relationship, the value of r will be close to 0.

Formula for the Correlation Coefficient r

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n(\sum x^2) - (\sum x)^2][n(\sum y^2) - (\sum y)^2]}}$$

where n is the number of data pairs.

Assumptions for the Correlation Coefficient

1. The sample is a random sample.
2. The data pairs fall approximately on a straight line and are measured at the interval or ratio level.
3. The variables have a joint normal distribution. (This means that given any specific value of x , the y values are normally distributed; and given any specific value of y , the x values are normally distributed.)

Computation of correlation coefficient for the above data

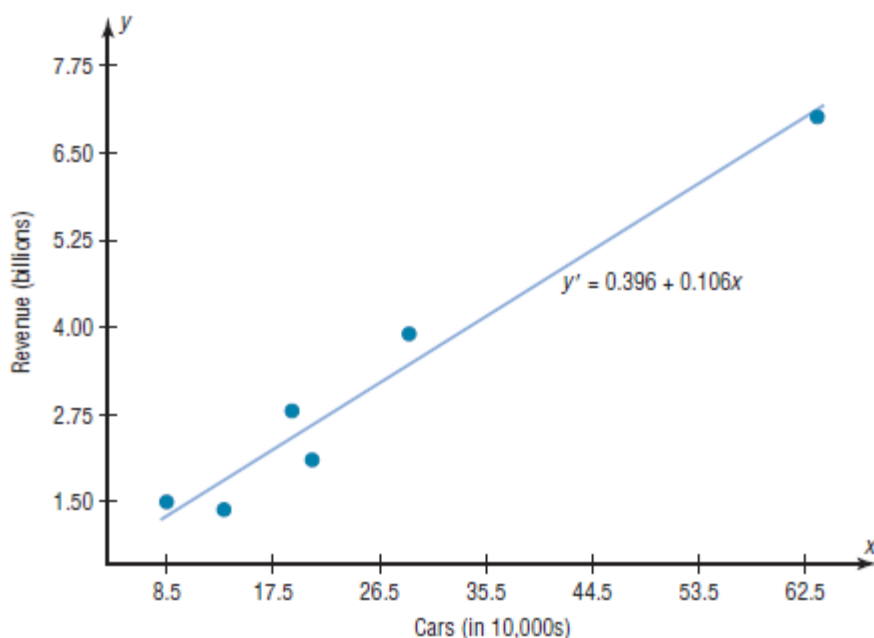
Company	Cars x (in 10,000s)	Revenue y (in billions)	xy	x^2	y^2
A	63.0	7.0	441.00	3969.00	49.00
B	29.0	3.9	113.10	841.00	15.21
C	20.8	2.1	43.68	432.64	4.41
D	19.1	2.8	53.48	364.81	7.84
E	13.4	1.4	18.76	179.56	1.96
F	8.5	1.5	12.75	72.25	2.25
	$\Sigma x = 153.8$	$\Sigma y = 18.7$	$\Sigma xy = 682.77$	$\Sigma x^2 = 5859.26$	$\Sigma y^2 = 80.67$

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Substitute in the formula and solve for r :

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n(\sum x^2) - (\sum x)^2][n(\sum y^2) - (\sum y)^2]}}$$
$$= \frac{(6)(682.77) - (153.8)(18.7)}{\sqrt{[(6)(5859.26) - (153.8)^2][(6)(80.67) - (18.7)^2]}} = 0.982$$

The correlation coefficient suggests a strong relationship between the number of cars a rental agency has and its annual revenue.



Use the equation of the regression line to predict the income of a car rental agency that has 200,000 automobiles.

Solution

Since the x values are in 10,000s, divide 200,000 by 10,000 to get 20, and then substitute 20 for x in the equation.

$$\begin{aligned} y' &= 0.396 + 0.106x \\ &= 0.396 + 0.106(20) \\ &= 2.516 \end{aligned}$$

Hence, when a rental agency has 200,000 automobiles, its revenue will be approximately \$2.516 billion.